

List of Experiments

Sr. no	Statements	Remarks
1	Installation of Open source JDK, setting up of environment variables, working with Java compiler and JVM, executing simple programs a) to show usage of print() and println() methods. b) Write a program to print command-line arguments and respective length of each command-line argument.	
2	a) Write a program to find factorial of a given number using (i) static method (ii) non-static method for computing factorial b) Declare a class called complex to represent a complex number. Write a program to add two complex numbers. Define all required methods.	
3	Declare a class to have a stack for storing int numbers. Define required methods and also main. The inputs are to be given through command-line arguments e.g.: At prompt>java stack_main push 6 push 4 pop push 0 pop Also comment on the effect of parseInt method being called without a try and catch block. How will you take care of stack underflow and stack overflow conditions?	
4	a) Declare a class to represent a node to store an int for a singly linked list. Write a complete program to implement a linear singly linked list. Define following methods: • Constructor(s) • Append • Traverse • Search for a node containing specified information • main b) Simple program to show usage of All shift operators. c) Simple program to show usage of methods of the Scanner class to read inputs from the keyboard. d) Write a program to add two matrices of int to produce a resultant matrix. Assume the size of the matrix as m x n. Read inputs from the keyboard using method(s) of class Scanner. [A] = [B] + [C]	
5	a) Implement the stack incorporating underflow and overflow conditions using exceptions. b) Declare a class called buffer having an int array of length n to store only positive nos. Define required methods and main to meet following: Read pairs of location (loc) and value (v) until 0 0 given	

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	Write into the array value v at location loc if, Location loc is valid, else raise and handle an exception, and if value v is positive, else raise and handle an exception (of different type), and location loc is empty, else raise and handle an exception (of different type) i. e. over-writing is permitted.	
6	a) Write a program to illustrate usage of user-defined exception handling. Exceptions are to be raised for different error codes and corresponding messages are to be displayed using same exception handler. b) Declare a class shape having two data members to represent two dimensions of geometric shapes. create two subclasses - rect and tri corresponding to rectangle and triangle. Define method calculate_area for both the classes for computing corresponding area of the geometric shapes. Implement without using dynamic method dispatch.	
7	a) Modify shape, rect and tri classes exercise to demonstrate dynamic method dispatch. b) Write a program to illustrate following characteristics of interfaces: - An interface can be implemented by many classes - A class can implement more than one interfaces - An interface reference can be used to refer to an object of a class that has implemented the interface. Also use interface reference given as a dummy argument. c) Write a program to use packages and create sub classes in different packages. Also check the accessibility of members.	
8	Write a multi-threaded program wherein n child threads are created. Each child thread prints odd numbers in a given range. it sleeps for 1 sec after printing 10 odd numbers. (e.g. 0th thread processes between 1 to 100, 1st thread between 101 to 200 and so on) Use method join to terminate main thread last. Also test various methods to get and set name of a thread, to change the priority etc.	
9	a) Write a complete multi-threaded program to meet following - There are two types of threads - produce and consumer. One of each type. The producer produces 5 int information by writing it into a shared buffer, one at a time. The consumer consumes and prints fresh information produced from the shared buffer. The shared buffer can store only one info at a given time. b) Enhance above program where there are 3 producers and 2 consumer threads. Each producer prod	
10	a) Write a multi-threaded program to find & print prime numbers up to n. Each thread except the last processes 100 numbers. The prime numbers are	

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	to be printed in ascending order. Do not use array and sorting for prime numbers. b) Write a simple applet program to test various methods of Graphics. c) Write a program to have two buttons in an applet and print a message when the button is clicked.	
11	a) Write a complete program to create a GUI for accepting two int nos to be given through two TextFields and to compute and display results of addition and subtraction of those two numbers in a third text field. There are three buttons – Add, Subtract and clear for doing respective tasks. Use two panels and layout of your choice. b) Draw a polygon using mouse clicks (either in a frame or in an applet) Clear button to clear the polygon.	
12	a) Implement a complete program to have a stack of strings. Provide GUI to read and display string through a text field and three buttons - to push, pop strings and the third to clear text field. Use Stack class of java.util. b) Write changes required to have a stack of Integer.	
13	Write following programs using the Generics and util package: a) Create an array-list to store strings. Use methods add, remove, sort and toArray. Also use for-each loop. b) Create a hashtable to store / get details of students. id nos are to be used as keys.	